

1. A. Let X and Y be two independent random variables. This means: $E[XY] = E[X]E[Y]$.
 The covariance of X and Y is given by $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$.
 Therefore, $\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0$.

B. Example: Let X be a standard normal random variable, and let $Y = X^2$.
 $E[X] = 0$ and $E[Y] = E[X^2] = 1$. $\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = E[X^3] - 0 \cdot 1 = 0$
 (since $E[X^3] = 0$ for a standard normal distribution). X and Y are uncorrelated, but they are not independent because Y is completely determined by X .

C. Example: Let X , Y , and Z be random variables defined as follows: Let A , B be independent fair coin flips taking values in $\{0,1\}$, and define: $X = A$, $Y = B$ and $Z = A \oplus B$.
 Each pair (X,Y) , (X,Z) , and (Y,Z) is independent. But: $X = Z \oplus Y$, $Y = Z \oplus X$, $Z = X \oplus Y$. Any one variable can be determined from the other two.

D. Let $z \in \mathbb{R}^d$ be any vector. Consider:

$$z^T C z = z^T \left(\sum_{i=1}^n x_i x_i^T \right) z = \sum_{i=1}^n z^T x_i x_i^T z = \sum_{i=1}^n (x_i^T z)^2$$

Each term $(x_i^T z)^2 \geq 0 \Rightarrow$ sum is non-negative.

2. A. Let X be a standard normal distribution in one dimension. We are asked to compute: $P(|X| \leq 1) = P(-1 \leq X \leq 1)$. The CDF of the standard normal distribution at $X = 1$ is $\Phi(1) = 0.8413$ and the CDF at $X = -1$ is $\Phi(-1) = 1 - \Phi(1) = 0.1587$. Thus: $P(-1 \leq X \leq 1) = \Phi(1) - \Phi(-1) = 0.8413 - 0.1587 = 0.6826$
 Therefore, for $d=1$, the probability is 0.6826.

B. Consider a d -dimensional standard normal random vector $X \sim N(0, I_d)$. Meaning $X = (X_1, X_2, \dots, X_d)$, with $X_i \sim N(0,1)$, independent. Then: $\|X\|^2 = \sum_{i=1}^d X_i^2$.
 $\|X\|^2$ follows a chi-squared distribution with d degrees of freedom.

$$E[\|X\|^2] = d, \quad \text{Var}[\|X\|^2] = 2d$$

As $d \rightarrow \infty$, $\|X\|^2$ becomes increasingly concentrated around d . using the CDF of chi-squared distribution: $\lim_{d \rightarrow \infty} P(\|X\|^2 \leq 1) = 0 \Rightarrow \lim_{d \rightarrow \infty} P(\|X\| \leq 1) = 0$.

3. Let X be a random vector with covariance matrix C . Consider a linear transformation $y = w^T X$ where w is a constant vector. $E[y] = E[w^T X] = w^T E[X]$. Let $\mu = E[X]$, then: $\text{Var}(y) = E[(y - E[y])^2] = E[(w^T X - w^T E[X])^2] = E[(w^T (X - \mu))^2] = E[w^T (X - \mu)(X - \mu)^T w] \stackrel{w \text{ is a constant}}{=} w^T E[(X - \mu)(X - \mu)^T] w = w^T C w$
4. A. Let $A \in \mathbb{R}^{n \times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. the eigenvalues of a matrix are the roots of its characteristic polynomial: $\det(A - \lambda I) = 0$. But for any square matrix M , we have the identity: $\det(M) = \det(M^T)$ so: $\det(A^T - \lambda I) = \det((A - \lambda I)^T) = \det(A - \lambda I)$ Thus, A and A^T have the same characteristic polynomial, and hence the same eigenvalues.

B. If a matrix is diagonalizable, it can be written as: $A = PDP^{-1}$ where D is a diagonal matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ and P is the matrix of eigenvectors. Assume A is diagonalizable then: $\det(A) = \det(PDP^{-1}) = \det(P) \det(D) \det(P^{-1}) = \det(P) \det(D) \det(P)^{-1} = \det(D)$. Since D is diagonal with entries $\lambda_1, \lambda_2, \dots, \lambda_n$, we have: $\det(D) = \prod_{i=1}^n \lambda_i$. Conclusion: $\det(A) = \prod_{i=1}^n \lambda_i$.

5. A. Differentiability: case 1: $w^T x \geq 1$, then $f(w, x) = 0$. This is a constant function, which is differentiable everywhere in this region.

Case 2: $w^T x < 1$, then $f(w, x) = 1 - w^T x$. This is a linear function, which is differentiable everywhere in this region.

case 3: at the boundary: $w^T x = 1$ the function change abruptly indicating a potential non-differentiable point.

Conclusion: the function differentiable at $\mathbb{R}^d \setminus \{x \in \mathbb{R}^d : w^T x = 1\}$

Gradient: case 1: $w^T x \geq 1$, then $f(w, x) = 0 \Rightarrow \nabla f(x) = 0$.

Case 2: $w^T x < 1$, then $f(w, x) = 1 - w^T x \Rightarrow$ Gradient with respect to x : $\nabla_x f = -w$

$$\nabla f_x(w, x) = \begin{cases} 0 & \text{if } w^T x \geq 1 \\ -w & \text{if } w^T x < 1 \end{cases}$$

B. This function is differentiable throughout $\mathbb{R}^d$ because each element in it is differentiable by itself: $w^T x$ is linear which is differentiable for every x .

$e^{-w^T x}$ is differentiable for every x and $1 + e^{-w^T x}$ is differentiable for every x .

Conclusion: the function differentiable at $\mathbb{R}^d$.

Gradient: Let $z = w^T x \Rightarrow f(x) = h(z) = \frac{1}{1+e^{-z}}$. Then: $\frac{d}{dz} h(z) = h(z)(1 - h(z))$

According the chain rule: $\nabla f(x) = \frac{df}{dz} \cdot \nabla z = h(z)(1 - h(z)) \cdot w$. Since $f(x) = h(z)$ we get: $\nabla f(x) = f(x)(1 - f(x)) \cdot w$.